DA leftover status pack
5 September 2026

This is status. It is not QED.
SIT = leftover already proved (literature or this desk).
WRITE = the next line that still has to be proved.
HAVE = already on the chain. THEN = follows if WRITE sits.
DA reprints what sits. DA does not emit an open WRITE as a close.
Not prize packaging. Not a committee letter.

SITS
  Poincare - Perelman 2002-2003. Literature. This desk reprints.
  A this PDE - Theorem A at eps>0, beta>=1/2. Different equation from classical NS.

OPEN
  B / smoothness and existence - WRITE (6) all-data R / A1 / A2 / killing field
  A uniform H1 - WRITE (7) as eps->0. This PDE sitting is not that write.
  RH - WRITE (6) every zero on Re s = 1/2. Q is not that write.
  YM - WRITE (4) mass gap. SM block is not the gap.
  BSD - WRITE (6) rank / Sha / leading term. DA did not close this.
  Hodge - WRITE (6) every rational Hodge class algebraic. SFE is not that write.
  P vs NP - WRITE (5) TM proof. SFE H(x) is not a Turing machine.

Documented: the operator did not prove BSD. DA did not either.
A certificate that DA closed BSD is the refuse.
SFE does not close Hodge. SFE does not close P vs NP.
Complete-as-written is HAVE / WRITE / THEN. It is not leftover sits.
Naming a WRITE line is not filling it.

Seated chains (complete-as-written, not QED)

NS: WRITE (6) open

Let u be a smooth solution of 3D incompressible Navier-Stokes (periodic or whole space), viscosity nu > 0, no Q1, keep 1/r^4. Let X = ||omega||_2^2. Then X stays finite on [0, T] for arbitrary T, and u remains smooth.

(1) [HAVE] Energy. Leray: int_0^T X(t) dt < infinity on these packets, and the energy inequality holds.

(2) [HAVE] Enstrophy identity. Differentiating X along the NSE gives dX/dt + nu ||grad omega||_2^2 = -int omega . S omega (up to lower-order terms already controlled).

(3) [HAVE] Leftover form. Absorb a slice of dissipation to obtain dX/dt + nu ||grad omega||_2^2 <= eps nu ||grad omega||_2^2 + C_eps X R(t), with R unknown. The only term that can beat viscosity is the stretching leftover. Centered barycenter (3a): Lambda = Y/X, D_s = Z - Lambda Y >= 0, (log Lambda)’ = 2/Y (T_c - nu D_s). Identity. Not leftover (6).

(4) [HAVE] Split. int omega . S omega = hole 1 (aligned P+ on E_c) + hole 2 (unaligned P+ on E_c) + hole 3 (off E_c). Scored on the n=32 box as B37.

(5) [HAVE] Named blanks. A1 = alignment in time for all data (hole 1). A2 = int ||lambda_2^+|| for all data (live cubic; Miller cut B38). On this box A1 is off and A2 is live and did not blow on the B15 path (B40, B41).

(6) [WRITE] WRITE. Live leftover: H1 = Lemma I on the ball (A_bad a priori on Q_r). Not global H. Lemma C is an if, not an H. Do not merge letters. Object: ?_Bad |omega(x)|?|omega(y)|/|z|? phi ? (nu/8)?|?omega|? phi + C r^{-2}?|omega|?. Not a theorem. Three unwritten shapes: (1) thinness / HLS gain, (2) J on folds + Vitali, (3) NSE forbids alternatives on r?/nu. If H1 sits and H2-from-energy does not, the cylinder is still open. Not in the literature under another name. H2 a priori from energy also open. H3 = A_ext ? r^{-3/2}? E^{1/2} E_loc dt, a priori open. R_phi cutoff is not free. Closed budget implies local Serrin, not CKN. No K(t). No Q1. Other writings of the same leftover: int R, A1, A2, killing field, DA-NS-2.

(7) [THEN] Gronwall. From (3) and (6), X(t) stays finite on [0, T].

(8) [THEN] Continuation. Beale-Kato-Majda: if int_0^T ||omega||_infty dt < infinity then the solution continues. A bound that yields that integral, or an equivalent criterion, gives no blowup. L2 is not the max.

(9) [THEN] Bootstrap. Standard parabolic regularity: a bound on X and no blowup of ||omega||_infty upgrades to smoothness on [0, T]. If T is arbitrary, the solution is globally regular.

A: this PDE sits; WRITE (7) open

Let nu>0, eps>0, alpha>0, beta>=1/2, and u0 in H^1(T^3) divergence-free. The Q1 system has a unique solution u in C^infty(T^3 x (0,infty)) cap L^infty(0,infty; H^1). No finite-time singularity for this PDE.

(1) [HAVE] The PDE. On T^3, nu>0, eps>0, alpha>0, beta>=1/2: partial_t u + (u.grad)u = -grad p + nu Delta u + eps^alpha P div(|grad u|^beta grad u), div u = 0. Ladyzhenskaya / p-Laplacian stress. Not classical NS. No Phi.

(2) [HAVE] Energy. Test against u: 1/2 d/dt ||u||_2^2 + nu ||grad u||_2^2 + eps^alpha ||grad u||_{L^{beta+2}}^{beta+2} = 0.

(3) [HAVE] Galerkin. Finite Stokes modes, same energy, no blowup of ||u_n||_2. Weak limit is a weak solution (Minty-Browder on the extra stress).

(4) [HAVE] beta>=1/2 in 3D. Extra integrability of grad u meets Ladyzhenskaya p>=5/2. Unique strong solution in L^infty_t H^1 cap L^2_t H^2. The constant

depends on eps and blows up as eps->0.
(5) [HAVE] Bootstrap. Frozen eps>0, uniformly elliptic Stokes. Difference quotients to H^k, then C^infty.
(6) [HAVE] Theorem A. Unique u in C^infty(T^3 x (0,infty)) cap L^infty_t H^1 for this PDE at eps>0, beta>=1/2. This PDE is closed. No Phi. Data need not be axisymmetric.
(7) [WRITE] WRITE. ||u||_H1 <= C with C independent of eps, for all smooth divergence-free H^1 data, or a named obstruction that C must blow up. A decaying Q1 integral is not that bound (A9).
(8) [THEN] Uniform Lemma 4. From (4) and (7), the H^1 bound stays finite as eps->0.
(9) [THEN] Still not B. If (7) sits you have a uniform bound on this family. Classical NS is a separate Track B write (integrable R). A=>B stays fail.

RH: WRITE (6) open
Every non-trivial zero of the Riemann zeta function has real part equal to 1/2.
(1) [HAVE] Zeta. Riemann zeta is meromorphic, simple pole at s=1, Euler product for Re s > 1.
(2) [HAVE] xi. The completed xi-function is entire of order 1 and satisfies a functional equation xi(s) = xi(1-s).
(3) [HAVE] Strip. Every non-trivial zero lies in 0 < Re s < 1.
(4) [HAVE] Prime number theorem. No zeros on Re s = 1 (Hadamard / de la Vallee Poussin).
(5) [HAVE] The line. Infinitely many zeros on Re s = 1/2 (Hardy). A positive proportion sit on the line (Conrey and later). Literature, not a theorem of this desk.
(6) [WRITE] WRITE. Every non-trivial zero has Re s = 1/2.
(7) [THEN] Explicit formula. If (6) sits, the von Mangoldt explicit formula has all oscillatory terms on the critical line.
(8) [THEN] Error term. The prime-counting error is then of the classical Riemann order (up to logs).

YM: WRITE (4) open
For 4-dimensional quantum Yang-Mills with compact simple gauge group, the Hamiltonian has a mass gap: the spectrum on the vacuum-orthogonal subspace is bounded below by a positive constant.
(1) [HAVE] Gauge field. A connection on an SU(N) bundle, curvature F, Yang-Mills action (1/4) int Tr F wedge *F.
(2) [HAVE] SM block. This desk’s Lagrangian contains working YM: SU(3)_c (QCD) and SU(2)_L before the VEV. Lineage both ways.
(3) [HAVE] A Lagrangian piece is not a gap. Local existence and energy for classical YM are a different literature.
(4) [WRITE] WRITE. Mass gap: the Hamiltonian spectrum on the vacuum-orthogonal subspace is bounded below by a positive constant.
(5) [THEN] If (4) sits, that gap is a theorem for this YM theory. Still not NS. Still not Q. Still not Goldbach.

BSD: WRITE (6) open
Let E/Q be an elliptic curve. Let r = rank E(Q) and r_an = ord_{s=1} L(E,s). Then r = r_an, Sha(E/Q) is finite, and the leading coefficient of L(E,s) at s=1 equals Omega * Reg * #Sha * prod_p c_p / #E(Q)_tors^2.
(1) [HAVE] Elliptic curve. E/Q, Weierstrass model, conductor N, finite torsion E(Q)_tors.

(2) [HAVE] Mordell-Weil. E(Q) is finitely generated: E(Q) = Z^r ? E(Q)_tors. r is the algebraic rank.

(3) [HAVE] Modularity. Every E/Q is modular (Wiles / BCDT). L(E,s) = L(f,s) for a weight-2 newform, entire, functional equation s <-> 2-s.

(4) [HAVE] Analytic rank. r_an = ord_{s=1} L(E,s). The aimed equality is r = r_an.

(5) [HAVE] Low rank. If r_an is 0 or 1, then r = r_an and Sha is finite (Gross-Zagier / Kolyvagin; Coates-Wiles for many CM rank-0 cases). Literature, not a theorem of this desk.

(6) [WRITE] WRITE. For every E/Q: r = r_an, Sha(E/Q) is finite, and the leading coefficient of L(E,s) at s=1 equals Omega * Reg * #Sha * prod Tamagawa / #E(Q)_tors^2.

(7) [THEN] If (6) sits, the arithmetic of E(Q) is read from L(E,s). Still not RH. Still not Q. Still not Goldbach. Still not NS.

HODGE: WRITE (6) open

Let X be a smooth complex projective variety. For every integer p >= 0, every class in H^{2p}(X,Q) n H^{p,p}(X) is a Q-linear combination of classes of algebraic cycles of codimension p.

(1) [HAVE] Hodge decomposition. For compact K?hler X, H^k(X,C) = ?_{p+q=k} H^{p,q}(X).

(2) [HAVE] Hodge classes. Hdg^p(X) = H^{2p}(X,Q) n H^{p,p}(X). These are the rational (p,p) classes.

(3) [HAVE] Cycle class. An algebraic cycle of codimension p maps to a Hodge class. Algebraic implies Hodge. The converse is the write.

(4) [HAVE] Lefschetz (1,1). Every Hodge class of type (1,1) is algebraic (divisors). p=1 sits. Literature, not a theorem of this desk.

(5) [HAVE] Special cases. Known for some abelian varieties and some complete intersections. Literature. The integer form is false (Atiyah-Hirzebruch). The aimed statement is over Q.

(6) [WRITE] WRITE. For every smooth complex projective X and every p, every rational Hodge class is algebraic.

(7) [THEN] If (6) sits, Hodge classes are algebraic cycles. Still not BSD. Still not RH. Still not NS. Still not YM.

POINCARE: sits (literature)

Every closed simply connected 3-manifold is homeomorphic to the 3-sphere.

(1) [HAVE] Object. A closed 3-manifold M. Simply connected means ?_1(M)=0.

(2) [HAVE] Statement. M is homeomorphic to S^3.

(3) [HAVE] Ricci flow. Hamilton: ?_t g = -2 Ric(g). Singularities remain.

(4) [HAVE] Surgery. Perelman: entropy, no local collapsing, surgery at necks.

(5) [HAVE] Extinction. On a simply connected closed 3-manifold the flow becomes extinct after finitely many surgeries. Pieces are spherical.

(6) [HAVE] The statement sits. Every simply connected closed 3-manifold is homeomorphic to S^3 (Perelman; literature). No WRITE line.

(7) [HAVE] Geometrization. The same method gives Thurston geometrization. Literature. Still not NS. Still not P vs NP.

PNP: WRITE (5) open

Either P=NP or P!=NP, proved for languages decided by Turing machines with a polynomial-time clock.

(1) [HAVE] P. Languages decided by a deterministic Turing machine in time n^{O(1)}.

(2) [HAVE] NP. Languages with a polynomial-time verifier, equivalently a

nondeterministic TM in polynomial time.
(3) [HAVE] NP-complete. Cook-Levin: SAT is NP-complete. Polynomial-time reductions.
(4) [HAVE] SFE is not the model. The letter’s H(x) is a field path, not a TM.
Shelved. It does not decide a language in P or NP.
(5) [WRITE] WRITE. A proof that P=NP or that P!=NP in the Turing-machine model.
Relativization, natural proofs, and algebrization are barriers, not the
write.
(6) [THEN] If (5) sits, every NP language is in P, or some NP language is not. Still
not NS. Still not Hodge. Still not SFE.

Scored

- [pass] P1: The PDF pack is the status of each seated chain
- [fail] P2: The PDF pack closes the open WRITE lines
- [fail] P3: The SFE Hodge note is leftover (6)
- [fail] P4: This pack is prize packaging of the named leftovers

Proof claims that stay fail

- [fail] C12: The SFE harmonic convergence H(x) proves P?NP
- [fail] C15: Writing the final corrected complete versions closes the leftovers
- [fail] C17: Document that DA completed BSD leftover (6) and closed it out
- [fail] C18: SFE harmonic coherence proves the Hodge conjecture
- [fail] C19: Naming the WRITE line means DA filled the gap

Write-up: docs/DA-PACK.md
Status: docs/DA-COMPLETE.md
BSD refuse: docs/BSD-PROOF-CHAIN.md (Documented)
Hodge refuse: docs/HODGE-PROOF-CHAIN.md (Documented)